What is Regression?

Regression is used when you want to predict a **continuous numerical value.**
The goal is to find the **relationship between input features and output value.**

Types of Regression

1️⃣ Linear Regression

Finds the best straight line through data points.

Formula: $y = mx + c$

- y = predicted value
- m = slope (how much y changes when x changes)
- x = input feature
- c = intercept (value of y when x is 0)

Real examples:

- Predict salary based on experience
- Predict house price based on size
- Predict weight based on height

When to use:

- Relationship between input and output is linear
- Output is continuous
- Simple and interpretable

2️⃣ Multiple Linear Regression

Same as linear regression but with **multiple input features.**

Formula: $y = m_1x_1 + m_2x_2 + m_3x_3 + c$

Real example:

- Predict house price based on size, location, number of rooms, age of house
- Predict salary based on experience, education, city, skills

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3️⃣ Polynomial Regression

When relationship between input and output is **not a straight line** but a curve.

Real example:

- Predicting growth of a plant over time

- Predicting COVID cases over time
- Speed of a car during acceleration

4️⃣ Ridge Regression

Linear regression with **regularization** to prevent overfitting.

Overfitting means model performs very well on training data but poorly on new data.

Ridge adds a penalty to large coefficients to keep the model simple.

5️⃣ Lasso Regression

Similar to Ridge but can **reduce some feature weights to exactly zero.**

This means Lasso automatically does **feature selection** — removes unimportant features.

Key Evaluation Metrics for Regression

Metric	Meaning	Good value
MAE — Mean Absolute Error	Average of absolute differences between predicted and actual	Lower is better
MSE — Mean Squared Error	Average of squared differences — penalizes large errors more	Lower is better
RMSE — Root Mean Squared Error	Square root of MSE — same unit as output	Lower is better
R2 Score	How well model explains variance in data — 1.0 is perfect	Closer to 1 is better

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How Model Learns — Gradient Descent

This is how regression finds the best line.

1. Start with random line
2. Calculate error — how wrong is the prediction
3. Adjust slope and intercept to reduce error
4. Repeat thousands of times
5. Eventually finds the best line with minimum error

Think of it like walking down a hill blindfolded. Every step you take the direction that goes downhill. Eventually you reach the bottom — which is minimum error.

Overfitting vs Underfitting

Underfitting:

- Model is too simple
- Performs poorly on both training and test data
- Like a student who didn't study at all

Overfitting:

- Model is too complex
- Performs very well on training data but poorly on new data
- Like a student who memorized answers but can't apply concepts

Good fit:

- Performs well on both training and test data
- Generalizes to new unseen data

Practice These

What is Linear Regression?

"Linear regression predicts a continuous output by finding the best fit line through data points. The line is defined by y equals mx plus c where m is the slope and c is the intercept. The model learns by minimizing the error between predicted and actual values using gradient descent."

What is the difference between Ridge and Lasso?

"Both are regularized versions of linear regression that prevent overfitting. Ridge adds a penalty to large coefficients but never removes them completely. Lasso can reduce coefficients to exactly zero which means it performs automatic feature selection by removing unimportant features."

What is R2 score?

"R2 score measures how well the model explains the variance in the data. A score of 1 means perfect prediction. A score of 0 means the model is no better than just predicting the mean. Negative R2 means the model is worse than predicting the mean."

What is overfitting?

"Overfitting happens when the model learns the training data too well including noise and random patterns. It performs very well on training data but poorly on new unseen data. We prevent it using regularization, cross validation, or reducing model complexity."